Eugenio D. Macias

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EDUCATION

Brown University *GPA: 4.0* Providence, RI | *May 2026*
B.S. Applied Mathematics–Computer Science **(Honors)**, *B.A. Mathematics*

United World College (UWC) – *IB Diploma* Mostar, Bosnia & Herzegovina | *May 2022*

EXPERIENCE

Brown University, *Karniadakis Lab* – **Undergraduate Research Assistant** Providence, RI | *Jan 2026 – Present*
(Python, PyTorch, PINNs, L-BFGS-B, SSBroyden, Adam, JAX) DRP-Cahn-Hilliard-PINN | Paper

- Benchmarked **self-scaled quasi-Newton** optimizers (**SSBroyden** variants, **L-BFGS-B**) vs. **Adam** for **PINNs** on the 2D **Cahn–Hilliard** 4th-order phase-field **PDE**; best $\varepsilon_{L_2} = 0.258$ on 64^2 grid (**SSBroyden2**)
- Ran 100+ experiments across 53 configurations and 3 seeds; analyzed **loss landscape geometry**, **convergence rates**, and curvature alignment of quasi-Newton **Hessian** approximations

Citi – **Quantitative Analyst Intern**: *Equity Derivatives* NYC |*Jun 2025 – Aug 2025*

- Automated **pricing diagnostics** in a **Python–C++** interface for **exotic derivatives** across EMEA, NAM, APAC books
- Validated live **fair values** and **Greeks** via **stochastic/local vol MC** and **PDE** solvers; accelerated runtime by **2.5x**
- Obtained **1st place** (540+ participants) in Citi's firmwide **ML Competition** and **2nd** in internal **market-making** challenge
- **LightGBM+DART+SHAP** with **Bayesian optimization** on 6M+ exotics notes; **Swish NNs** achieving lowest **MSE**

Brown University – **Undergraduate Teaching Assistant** Providence, RI |*Aug 2025 – Present*

- *APMA 1200 – Operations Research* (Jan 2026–Present): **Markov chains**, **Poisson processes**, **Martingales**, **Brownian motion**
- *MATH 100 – Calculus II* (Aug–Dec 2025): integration, sequences & series, **Taylor expansions**, **multivariable calculus**

D. E. Shaw & Co. – *Latitude Fellowship* NYC |*Apr 2024 – May 2024*

- Built airline sector revenue forecast via **multivariate time-series regression** in **Python**; achieved $R^2 = 0.87$ on hold-out test set
- Explored **alpha-generating signals** and **execution latency** effects for **equities** and **ETFs**

Buddywise AI – *Founder's Associate Intern* Berlin, Germany |*May 2023 – Jul 2023*

- Researched and optimized datasets for **ML pipelines** in safety-critical **computer vision**; reduced processing time by **35%**
- Secured $50,000 in Mixpanel credits; improved **A/B testing** and UI **KPI** visibility by **40%**

PROJECTS

Projects

Honors Senior Thesis: ***Weak-Form PDE Discovery for Stochastic Collective Dynamics*** *(Python, PyTorch, JAX, SciPy)*
Advisor: Prof. Bjorn Sandstede wsindy-manifold | Thesis

- Applied **WSINDy** to discover interpretable coupled PDEs for macroscopic density evolution from noisy Vicsek-alignment + Morse attraction–repulsion trajectories; sparsity via MSTLS, integration via ETDRK4
- Benchmarked against equation-free ROMs: FFT-aligned snapshots compressed by POD, evolved in latent space by **MVAR** and **LSTM**; evaluated across 7 dynamical regimes, 4 IC families, and multiple noise levels

RL Research: Curiosity-Driven Transfer in ProcGen *(Python, PyTorch, PPO, ICM, RND, CFN)*
Advisor: Prof. George Konidaris TransferRL | Paper

- Factorial sweep over four intrinsic signals (ICM, RND, CFN, NLL-Surprise) augmenting **PPO** with IMPALA-CNN on **ProcGen**; normalization (clip/RMS) and $\eta \in \{0.001 - -0.05\}$ as primary experimental variables
- Zero-shot and fine-tuned transfer; dynamics-aware vs. coverage-only signals, 40% cutoff, VLM-guided (CLIP) reward

Hamiltonian & Symplectic NNs for Charged Particle Dynamics *(PINNs, SympNets, PyTorch)* Paper

- Poisson NNs with PDE residual loss learn Hamiltonian structure in EM fields; sub-0.007 MSE, 2% parameter recovery
- SympNets via Lie-Darboux transforms learn symplectomorphisms from trajectories, preserving phase-space volume

Market-Neutral Long-Short Equity Strategy *(Python, Scikit-Learn, Pandas)* Paper

- PCA factor decomposition on 5,000 equities; OLS idiosyncratic alpha residuals; dollar-neutral long-short top/bottom quintile; Sharpe 2.12 out-of-sample

Research Paper: Fourier Analysis and Laplace's Equation *(LaTeX)* Paper

- Fourier series for Laplace's 2D BVP; four Dirichlet configs; L^2 convergence & truncation error analysis (50 pp.)

AWARDS

- **Shelby Davis Scholar**: Full tuition, Brown
- **Sprint Award**: $7,000 research internship grant
- **Citadel** Invitational Terminal 2024
- **Point72 & Correlation1**: Data Scholar 2024
- **Sigma Xi**: Invited Member, Research Honor Society
- **Bridgewater Associates**: Investment Immersion 2024
- **PUTNAM 2024**: top 20% nationwide
- **IYMC**: Silver Medalist, Mathematics

SKILLS & INTERESTS

Math Notes

Skills: Exceptional quantitative intuition, rapid mental arithmetic & probabilistic reasoning; **AI-augmented** research workflows; competitive chess (**2070** ELO, top 1%); strong cross-disciplinary pattern recognition
Languages: Spanish (**Native**), English (**Native**), German (**B1** — Intermediate), French (**A2** — Elementary)
Interests: Classical Guitar (**3rd** National, Mexico), Running (**5** marathons), Travel (**45** countries), Asian food